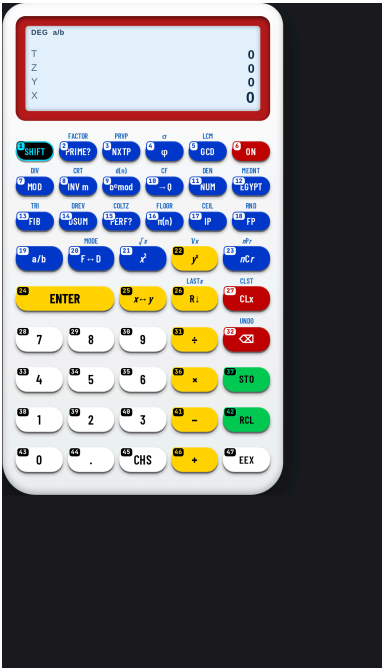
U-45 CALCULATORS

User Manual

Calculator #19 Exact-Fraction RPN Scientific (walnut)

Scientific – RPN



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#19 — Exact-Fraction RPN Scientific (walnut)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 SHIFT]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

four-level RPN with an exact rational (fraction) stack: key a number, ENTER, key the next, then the operation.

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
6. Worked examples
7. Tricks & hacks
8. Keyboard shortcuts
9. Common mistakes
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12. Appendix B — Function finder (A–Z)
13. Appendix C — Glossary
14. Appendix D — Model notes & limits
15. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#19** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#28, #29, #30, #33, #34, #35, #38, #39, #40, #43, #44, #45, #47
op	Operators & execute	#22, #24, #25, #26, #31, #36, #41, #46
fn	Functions	#2, #3, #4, #5, #7, #8, #9, #10, #11, #12, #13, #14, #15, #16, #17, #18, #19, #20, #21, #23
mem	Memory, variables & registers	#37, #42
clr	Clear, delete & power	#6, #27, #32

mod	**Shift, modes, menus & navigation**	**#1**

2 · Features

45 features: 31 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Exact rational stack: every level holds a fraction, never a rounded decimal**
2. LIVE **Fraction entry with the a/b key (3 a/b 4)**
3. LIVE **Decimal entry converts exactly (0.75 → 3/4)**
4. LIVE **Display cycle: improper fraction → mixed number → decimal**
5. LIVE **Four-level stack X Y Z T with ENTER, x↔y, R↓, LAST X and UNDO**
6. LIVE **→Q best fraction approximation (denominator ≤ 10 000)**
7. LIVE **NUM and DEN extract numerator and denominator**
8. LIVE **GCD and LCM**
9. LIVE **MOD and integer division DIV**
10. LIVE **Primality test IS PRIME?**
11. LIVE **Prime factorization screen with exponents**
12. LIVE **Next prime and previous prime**
13. LIVE **Euler totient φ(n)**
14. LIVE **Divisor sum σ(n) and divisor count d(n)**
15. LIVE **Modular inverse a^{-1} mod m**
16. LIVE **Modular power b^e mod m (BigInt, any size exponent)**
17. LIVE **Chinese remainder theorem for two congruences**
18. LIVE **Fibonacci F(n)**
19. LIVE **Digit sum and digit reversal**

20. LIVE Perfect / abundant / deficient classification
21. LIVE Continued-fraction expansion
22. LIVE Egyptian-fraction (greedy) decomposition
23. LIVE Mediant of two fractions
24. LIVE FLOOR, CEIL, ROUND, IP and FP
25. LIVE nCr and nPr
26. LIVE Triangular numbers
27. LIVE Collatz step count
28. LIVE Prime-counting π(n)
29. LIVE Exact powers of fractions with integer exponents
30. LIVE Exact square roots of perfect-square fractions
31. LIVE Registers R0–R9 via STO / RCL + digit
32. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
33. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
34. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
35. APP/UI Show / hide key numbers with one button (remembered between sessions)
36. APP/UI Copy the current display text to the clipboard
37. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
38. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
39. APP/UI Reference photo viewer and tool-tip name on every key

40. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
41. PLANNED **Stack history view**
42. PLANNED **Undo**
43. PLANNED **Equation library**
44. PLANNED **Speech read-out**
45. PLANNED **Colour-blind key theme**

2b · Feature guide

Calculator #19 is a four-level RPN with an exact rational (fraction) stack: key a number, ENTER, key the next, then the operation. Each card below is one of its 31 working features, with the keys that reach it and, where one exists, a verified example.

F1 Exact rational stack: every level holds a fraction, never a rounded decimal

Where: `#20` F↔D Fraction / mixed / decimal display; `#10` →Q Best fraction approximation; `#12` EGYPT Egyptian fraction

See it in action: Numerator of a decimal — *0.75 NUM* = 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Fraction entry with the a/b key (3 a/b 4)

Where: `#19` a/b Fraction bar during entry; `#10` →Q Best fraction approximation; `#12` EGYPT Egyptian fraction

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 Decimal entry converts exactly (0.75 → 3/4)

Where: #19 a/b Fraction bar during entry; #20 F↔D Fraction / mixed / decimal display; #24 ENTER ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value

See it in action: Numerator of a decimal — *0.75 NUM* = 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Display cycle: improper fraction → mixed number → decimal

Where: #20 F↔D Fraction / mixed / decimal display; #44 . Decimal point. Pressing it first gives "0."; a second point in the same number is ignored; #10 →Q Best fraction approximation

See it in action: Numerator of a decimal — *0.75 NUM* = 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Four-level stack X Y Z T with ENTER, x ↔ y, R↓, LAST X and UNDO

Where: #24 ENTER ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value; #26 R↓ R↓ — roll the stack down; #32 ⌫ Backspace — delete the last digit typed

See it in action: Add two fractions — *3/4 ENTER 1/6 +* = 0.91666667; Greatest common divisor — *12 ENTER 18 GCD* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 →Q best fraction approximation (denominator ≤ 10 000)

Where: #10 →Q Best fraction approximation; #11 NUM Numerator; #12 EGYPT Egyptian fraction

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 NUM and DEN extract numerator and denominator

Where: #11 NUM **Numerator**

See it in action: Numerator of a decimal — *0.75 NUM* = 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 GCD and LCM

Where: #5 GCD **GCD of Y and X**

See it in action: Greatest common divisor — *12 ENTER 18 GCD* = 6; Least common multiple — *4 ENTER 6 LCM* = 12

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 MOD and integer division DIV

Where: #7 MOD Y mod X; #8 INV m Y^{-1} mod X; #9 b^emod Z^Y mod X

See it in action: Modular power — *3 ENTER 4 ENTER 5 b^emod* = 1; Modular inverse — *3 ENTER 7 INV mod* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Primality test IS PRIME?

Where: #2 PRIME? Primality test of X; #3 NXTP Next prime

See it in action: Next prime — *NEXTP after 14* = 17

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Prime factorization screen with exponents

Where: #2 PRIME? Primality test of X; #3 NXTP Next prime

See it in action: Next prime — *NEXTP after 14* = 17

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 Next prime and previous prime

Where: #3 NXTP Next prime; #2 PRIME? Primality test of X; #1 SHIFT SHIFT / 2nd — the next key uses its upper (shift-coloured) function

See it in action: Next prime — *NEXTP after 14* = 17

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Euler totient φ(n)

Where: #4 φ Euler totient

See it in action: Euler totient — *φ(10)* = 4

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 Divisor sum σ(n) and divisor count d(n)

Where: #4 φ Euler totient; #9 b^e mod Z^Y mod X; #14 DSUM Digit sum

See it in action: Greatest common divisor — *12 ENTER 18 GCD* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Modular inverse a^{-1} mod m

Where: #7 MOD Y mod X; #8 INV m Y^{-1} mod X; #9 b^emod Z^Y mod X

See it in action: Modular power — *3 ENTER 4 ENTER 5 b^emod* = 1; Modular inverse — *3 ENTER 7 INV mod* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Modular power b^e mod m (BigInt, any size exponent)

Where: #22 yx y^x — raise Y to the power X (RPN: base first, then exponent); #47 EEX EEX — enter a power-of-ten exponent for the number in X; #6 ON ON — power on

See it in action: Modular power — *3 ENTER 4 ENTER 5 b^emod* = 1; Modular inverse — *3 ENTER 7 INV mod* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Chinese remainder theorem for two congruences

Where: #8 INV m Y^{-1} mod X

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 Fibonacci F(n)

Where: #13 FIB Fibonacci number

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 Digit sum and digit reversal

Where: #14 DSUM Digit sum; #24 ENTER ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value; #28 7 Digit 7

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Perfect / abundant / deficient classification

Where: #15 PERF? Perfect / abundant / deficient

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 Continued-fraction expansion

Where: #10 →Q Best fraction approximation; #12 EGYPT Egyptian fraction; #19 a/b Fraction bar during entry

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Egyptian-fraction (greedy) decomposition

Where: #12 EGYPT Egyptian fraction; #10 →Q Best fraction approximation; #19 a/b Fraction bar during entry

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 Mediant of two fractions

Where: #12 EGYPT Egyptian fraction

See it in action: Add two fractions — *3/4 ENTER 1/6 +* = 0.91666667

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 FLOOR, CEIL, ROUND, IP and FP

Where: #16 π(n) Count primes ≤ n; #17 IP ip; #18 FP fp

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 nCr and nPr

Where: #23 nCr Combinations nCr

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Triangular numbers

Where: #13 FIB Fibonacci number

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Collatz step count

Where: #9 b^e mod Z^Y mod X; #15 PERF? Perfect / abundant / deficient; #16 π(n) Count primes ≤ n

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Prime-counting π(n)

Where: #2 PRIME? Primality test of X; #3 NXTP Next prime

See it in action: Next prime — *NEXTP after 14* = 17

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Exact powers of fractions with integer exponents

Where: #7 MOD Y mod X

See it in action: Add two fractions — *3/4 ENTER 1/6 +* = 0.91666667

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Exact square roots of perfect-square fractions

Where: #21 x^2 Square x^2; #15 PERF? Perfect / abundant / deficient

See it in action: Add two fractions — *3/4 ENTER 1/6 +* = 0.91666667

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F31 Registers R0–R9 via STO / RCL + digit

Where: #14 DSUM Digit sum; #24 ENTER ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value; #28 7 Digit 7

Try it: Press Reset, use the key(s) above, and write the display here:

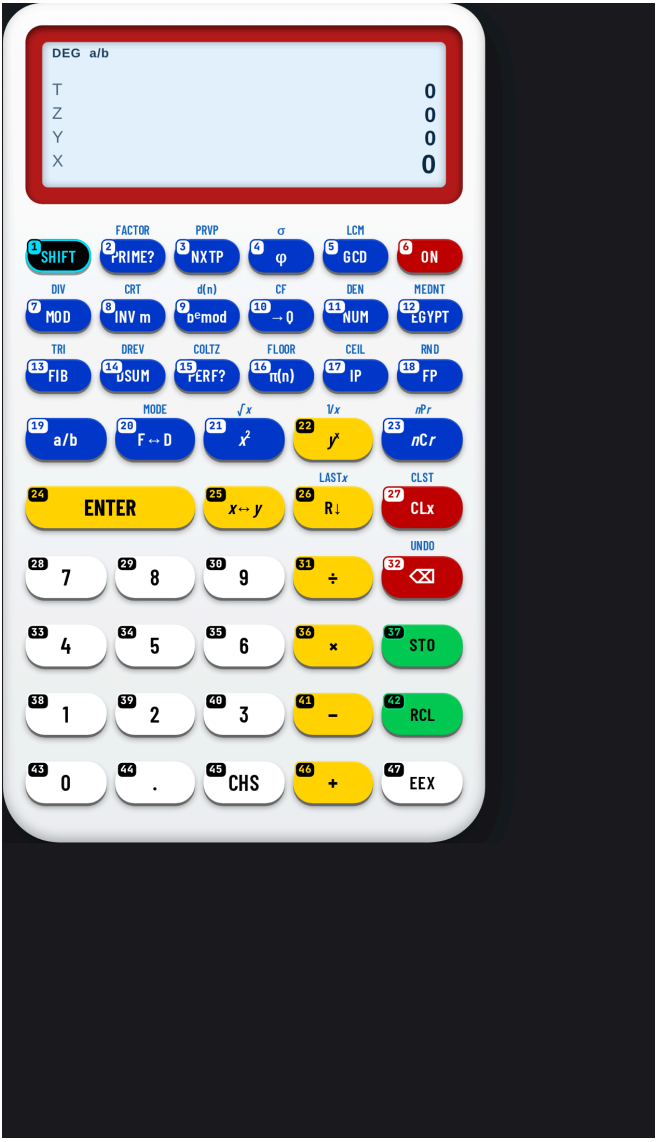
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #19 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	SHIFT	·	·	·	mod
2	PRIME?	FACTOR	·	·	fn
3	NXTP	PRVP	·	·	fn
4	φ	σ	·	·	fn
5	GCD	LCM	·	·	fn
6	ON	·	·	·	clr
7	MOD	DIV	·	·	fn
8	INV m	CRT	·	·	fn
9	b^e mod	d(n)	·	·	fn
10	→Q	CF	·	·	fn
11	NUM	DEN	·	·	fn
12	EGYPT	MEDNT	·	·	fn
13	FIB	TRI	·	·	fn
14	DSUM	DREV	·	·	fn
15	PERF?	COLTZ	·	·	fn
16	π(n)	FLOOR	·	·	fn
17	IP	CEIL	·	·	fn
18	FP	RND	·	·	fn
19	a/b	·	·	·	fn
20	F↔D	MODE	·	·	fn
21	x^2	√x	·	·	fn
22	yx	1/x	·	·	op
23	nCr	nPr	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	ENTER	·	·	·	op
25	x ↔ y	·	·	·	op
26	R↓	LASTx	·	·	op
27	CLx	CLST	·	·	clr
28	7	·	·	·	num
29	8	·	·	·	num
30	9	·	·	·	num
31	÷	·	·	·	op
32	⌫	UNDO	·	·	clr
33	4	·	·	·	num
34	5	·	·	·	num
35	6	·	·	·	num
36	×	·	·	·	op
37	STO	·	·	·	mem
38	1	·	·	·	num
39	2	·	·	·	num
40	3	·	·	·	num
41	−	·	·	·	op
42	RCL	·	·	·	mem
43	0	·	·	·	num
44	.	·	·	·	num
45	CHS	·	·	·	num
46	+	·	·	·	op
47	EEX	·	·	·	num

4 · Key-by-key reference

Every key on calculator #19, in number order, with every function it carries in this app.

#1 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#2 PRIME? `Functions`

Primary: Primality test of X

shifted (2nd / f / SHIFT layer) (press #1 first): FACTOR — Prime factorization

#3 NXTP `Functions`

Primary: Next prime

shifted (2nd / f / SHIFT layer) (press #1 first): PRVP — Previous prime

#4 φ `Functions`

Primary: Euler totient

shifted (2nd / f / SHIFT layer) (press #1 first): σ — Divisor sum

#5 GCD `Functions`

Primary: GCD of Y and X

shifted (2nd / f / SHIFT layer) (press #1 first): LCM — LCM of Y and X

#6 ON Clear, delete & power

Primary: ON — power on

#7 MOD Functions

Primary: Y mod X

shifted (2nd / f / SHIFT layer) (press #1 first): DIV — Integer quotient

#8 INV m Functions

Primary: Y^{-1} mod X

shifted (2nd / f / SHIFT layer) (press #1 first): CRT — Chinese remainder (T mod Z, Y mod X)

#9 b^emod Functions

Primary: Z^Y mod X

shifted (2nd / f / SHIFT layer) (press #1 first): d(n) — Divisor count

#10 → Q Functions

Primary: Best fraction approximation

shifted (2nd / f / SHIFT layer) (press #1 first): CF — Continued fraction

#11 NUM Functions

Primary: Numerator

shifted (2nd / f / SHIFT layer) (press #1 first): DEN — Denominator

#12 EGYPT Functions

Primary: Egyptian fraction

shifted (2nd / f / SHIFT layer) (press #1 first): MEDNT — Mediant

#13 FIB Functions

Primary: Fibonacci number

shifted (2nd / f / SHIFT layer) (press #1 first): TRI — Triangular number

#14 DSUM Functions

Primary: Digit sum

shifted (2nd / f / SHIFT layer) (press #1 first): DREV — Reverse digits

#15 PERF? Functions

Primary: Perfect / abundant / deficient

shifted (2nd / f / SHIFT layer) (press #1 first): COLTZ — Collatz steps

#16 π(n) Functions

Primary: Count primes ≤ n

shifted (2nd / f / SHIFT layer) (press #1 first): FLOOR — floor

#17 IP Functions

Primary: ip

shifted (2nd / f / SHIFT layer) (press #1 first): CEIL — ceil

#18 FP Functions

Primary: fp

shifted (2nd / f / SHIFT layer) (press #1 first): RND — Round to the displayed precision

#19 a/b Functions

Primary: Fraction bar during entry

#20 F ↔ D Functions

Primary: Fraction / mixed / decimal display

shifted (2nd / f / SHIFT layer) (press #1 first): MODE — SETUP — angle unit, number format, display options

#21 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #1 first): $\sqrt{}$x — Square root $\sqrt{}$

Keyboard: q

#22 yx Operators & execute

Primary: y^x — raise Y to the power X (RPN: base first, then exponent)

shifted (2nd / f / SHIFT layer) (press #1 first): 1/x — Reciprocal x^{-1} (1/x)

Keyboard: ^

#23 nCr Functions

Primary: Combinations nCr

shifted (2nd / f / SHIFT layer) (press #1 first): nPr — Permutations nPr

#24 ENTER Operators & execute

Primary: ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value

Keyboard: Enter

#25 x ↔ y Operators & execute

Primary: x≷y — exchange X and Y

#26 R↓ Operators & execute

Primary: R↓ — roll the stack down

shifted (2nd / f / SHIFT layer) (press #1 first): LASTx — LASTx — recall the X value used by the last operation

#27 CLx Clear, delete & power

Primary: CLx — clear X

shifted (2nd / f / SHIFT layer) (press #1 first): CLST — Clear stack

#28 **7** `Digits & number entry`

Primary: Digit 7

Keyboard: `7`

#29 **8** `Digits & number entry`

Primary: Digit 8

Keyboard: `8`

#30 **9** `Digits & number entry`

Primary: Digit 9

Keyboard: `9`

#31 **÷** `Operators & execute`

Primary: Division ÷

Keyboard: `/`

#32 ⌫ `Clear, delete & power`

Primary: Backspace — delete the last digit typed

shifted (2nd / f / SHIFT layer) (press #1 first): UNDO — undo

Keyboard: `Backspace` `Delete`

#33 4 Digits & number entry

Primary: Digit 4

Keyboard: 4

#34 5 Digits & number entry

Primary: Digit 5

Keyboard: 5

#35 6 Digits & number entry

Primary: Digit 6

Keyboard: 6

#36 × Operators & execute

Primary: Multiplication ×

Keyboard: * x

#37 STO Memory, variables & registers

Primary: STO — store the value into a memory / variable (press the target next)

#38 1 Digits & number entry

Primary: Digit 1

Keyboard: 1

#39 2 Digits & number entry

Primary: Digit 2

Keyboard: 2

#40 3 Digits & number entry

Primary: Digit 3

Keyboard: 3

#41 – Operators & execute

Primary: Subtraction –

Keyboard: -

#42 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

#43 0 Digits & number entry

Primary: Digit 0

Keyboard: 0

#44 . Digits & number entry

Primary: Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

Keyboard: . ,

#45 CHS `Digits & number entry`

Primary: Change sign (+/−) of the number being typed or the current result

#46 + `Operators & execute`

Primary: Addition +

Keyboard: +

#47 EEX `Digits & number entry`

Primary: EEX — enter a power-of-ten exponent for the number in X

Keyboard: e

5 · Worked examples

8 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Add two fractions

Calculate 3/4 ENTER 1/6 +. Press Reset first.

#40 3 → #19 a/b → #33 4 → #24 ENTER → #38 1 → #19 a/b → #35 6 → #46 +

Display shows X11/12 — expected 0.91666667. ✔ verified on the app.

Example 5.2 — Greatest common divisor

Calculate 12 ENTER 18 GCD. Press Reset first.

#38 1 → #39 2 → #24 ENTER → #38 1 → #29 8 → #5 GCD

Display shows X6 — expected 6. ✔ verified on the app.

Example 5.3 — Least common multiple

Calculate 4 ENTER 6 LCM. Press Reset first.

#33 4 → #24 ENTER → #35 6 → #1 SHIFT → #5 LCM

Display shows X12 — expected 12. ✔ verified on the app.

Example 5.4 — Euler totient

Calculate **φ(10)**. Press **Reset** first.

#38 1 → **#43** 0 → **#4** φ

Display shows X4 — expected 4. ✔ verified on the app.

Example 5.5 — Next prime

Calculate **NEXTP after 14**. Press **Reset** first.

#38 1 → **#33** 4 → **#3** NXTP

Display shows X17 — expected 17. ✔ verified on the app.

Example 5.6 — Modular power

Calculate **3 ENTER 4 ENTER 5 b^e mod**. Press **Reset** first.

#40 3 → **#24** ENTER → **#33** 4 → **#24** ENTER → **#34** 5 → **#9** b^e mod

Display shows X1 — expected 1. ✔ verified on the app.

Example 5.7 — Modular inverse

Calculate **3 ENTER 7 INV mod**. Press **Reset** first.

#40 3 → **#24** ENTER → **#28** 7 → **#8** INV m

Display shows X5 — expected 5. ✔ verified on the app.

Example 5.8 — Numerator of a decimal

Calculate 0.75 NUM. Press Reset first.

#43 0 → #44 . → #28 7 → #34 5 → #11 NUM

Display shows X3 **— expected 3. ✔ verified on the app.**

6 · Tricks & hacks

Hack 1: No equals key

RPN works number, number, operator: 3 ENTER 4 + = 7. Results stay on the stack for the next step.

Hack 2: Chain without ENTER

After an operation, just type the next number and operator: 3 ENTER 4 + 5 × gives (3+4)×5 = 35.

Hack 3: Swap to fix order

If you entered numbers in the wrong order for − ÷ or y^x, press x≷y before the operator.

Hack 4: LASTx to reuse

LASTx brings back the X value of the previous operation — useful for a × a.

Hack 5: Stack as memory

Up to four values are kept; R↓ rolls through them.

Hack 6: Constant via stack top

On 4-level stacks the T register copies down, so a constant multiplier can be reused by pressing × repeatedly.

Hack 7: Hidden functions on this model

23 extra functions sit on shifted layers. Press the layer key once, then the key: for example #1 then #2 → FACTOR; #1 then #3 → PRVP; #1 then #4 → σ; #1 then #5 → LCM.

Hack 8: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #47.

Hack 9: Type faster on a computer

20 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 10: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 11: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 12: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
q	#21	x^2	Square x^2
^	#22	yx	y^x — raise Y to the power X (RPN: base first, then exponent)
Enter	#24	ENTER	ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value
7	#28	7	Digit 7
8	#29	8	Digit 8
9	#30	9	Digit 9
/	#31	÷	Division ÷
Backspace Delete	#32	⌫	Backspace — delete the last digit typed
4	#33	4	Digit 4
5	#34	5	Digit 5
6	#35	6	Digit 6
* x	#36	×	Multiplication ×
1	#38	1	Digit 1
2	#39	2	Digit 2
3	#40	3	Digit 3
-	#41	–	Subtraction –
0	#43	0	Digit 0
. ,	#44	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
+	#46	+	Addition +
e	#47	EEX	EEX — enter a power-of-ten exponent for the number in X

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Pressing ENTER after an operation

ENTER after + duplicates the result. Usually you want to type the next number directly.

Mistake 2: Wrong order for – and ÷

Y – X and Y ÷ X: enter the first operand first. 10 ENTER 4 – = 6.

Mistake 3: CHS vs –

Make a number negative with CHS/+/–, not with –.

Mistake 4: Prefix left active

If f / g / SHIFT is shown, the next key is shifted. Clear it with the prefix-clear function.

Mistake 5: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 6: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (8)

Work each one on calculator #19, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 3/4 ENTER 1/6 + = _____
2. 12 ENTER 18 GCD = _____
3. 4 ENTER 6 LCM = _____
4. φ(10) = _____
5. NEXTP after 14 = _____
6. 3 ENTER 4 ENTER 5 b^emod = _____
7. 3 ENTER 7 INV mod = _____
8. 0.75 NUM = _____

Set B — Key-finding drills (59)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “SHIFT” (SHIFT / 2nd)? _____
2. Which key is “PRIME?” (Primality test of X)? _____
3. Which key is “NXTP” (Next prime)? _____
4. Which key is “φ” (Euler totient)? _____
5. Which key is “GCD” (GCD of Y and X)? _____
6. Which key is “ON” (ON)? _____

7. Which key is “MOD” (Y mod X)? _____

8. Which key is “INV m” (Y^{-1} mod X)? _____

9. Which key is “b^emod” (Z^Y mod X)? _____

10. Which key is “ → Q” (Best fraction approximation)? _____

11. Which key is “NUM” (Numerator)? _____

12. Which key is “EGYPT” (Egyptian fraction)? _____

13. Which key is “FIB” (Fibonacci number)? _____

14. Which key is “PERF?” (Perfect / abundant / deficient)? _____

15. Which key is “π(n)” (Count primes ≤ n)? _____

16. Which key is “IP” (ip)? _____

17. Which key is “FP” (fp)? _____

18. Which key is “a/b” (Fraction bar during entry)? _____

19. Which key is “F ↔ D” (Fraction / mixed / decimal display)? _____

20. Which key is “x^2” (Square x^2)? _____

21. Which key is “yx” (y^x)? _____

22. Which key is “nCr” (Combinations nCr)? _____

23. Which key is “ENTER” (ENTER)? _____

24. Which key is “x ↔ y” (x≷y)? _____

25. Which key is “R↓” (R↓)? _____

26. Which key is “CLx” (CLx)? _____

27. Which key is "÷" (Division ÷)? _____

28. Which key is "⌫" (Backspace)? _____

29. Which key is "×" (Multiplication ×)? _____

30. Which key is "STO" (STO)? _____

31. Which key is "−" (Subtraction −)? _____

32. Which key is "RCL" (RCL)? _____

33. Which key is "." (Decimal point. Pressing it first gives "0."; a second point in the same number is ignored)? _____

34. Which key is "CHS" (Change sign)? _____

35. Which key is "+" (Addition +)? _____

36. Which key is "EEX" (EEX)? _____

37. How do you reach "FACTOR"? _____

38. How do you reach "PRVP"? _____

39. How do you reach "σ"? _____

40. How do you reach "LCM"? _____

41. How do you reach "DIV"? _____

42. How do you reach "CRT"? _____

43. How do you reach "d(n)"? _____

44. How do you reach "CF"? _____

45. How do you reach "DEN"? _____

46. How do you reach "MEDNT"? _____

47. How do you reach “TRI”? _____

48. How do you reach “DREV”? _____

49. How do you reach “COLTZ”? _____

50. How do you reach “FLOOR”? _____

51. How do you reach “CEIL”? _____

52. How do you reach “RND”? _____

53. How do you reach “MODE”? _____

54. How do you reach “√x”? _____

55. How do you reach “1/x”? _____

56. How do you reach “nPr”? _____

57. How do you reach “LASTx”? _____

58. How do you reach “CLST”? _____

59. How do you reach “UNDO”? _____

Appendix A — Drill answers

Set A

1. **0.91666667** (display `X11/12`)

   `#40` 3 → `#19` a/b → `#33` 4 → `#24` ENTER → `#38` 1 → `#19` a/b → `#35` 6 → `#46` +

2. **6** (display `X6`)

   `#38` 1 → `#39` 2 → `#24` ENTER → `#38` 1 → `#29` 8 → `#5` GCD

3. **12** (display `X12`)

   `#33` 4 → `#24` ENTER → `#35` 6 → `#1` SHIFT → `#5` LCM

4. **4** (display `X4`)

   `#38` 1 → `#43` 0 → `#4` φ

5. **17** (display `X17`)

   `#38` 1 → `#33` 4 → `#3` NXTP

6. **1** (display `X1`)

   `#40` 3 → `#24` ENTER → `#33` 4 → `#24` ENTER → `#34` 5 → `#9` b^emod

7. **5** (display `X5`)

   `#40` 3 → `#24` ENTER → `#28` 7 → `#8` INV m

8. **3** (display `X3`)

   `#43` 0 → `#44` . → `#28` 7 → `#34` 5 → `#11` NUM

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #15
15. #16
16. #17
17. #18
18. #19
19. #20
20. #21
21. #22
22. #23
23. #24
24. #25
25. #26
26. #27
27. #31
28. #32
29. #36
30. #37
31. #41
32. #42
33. #44
34. #45
35. #46
36. #47
37. #1 SHIFT then #2 PRIME?
38. #1 SHIFT then #3 NXTP
39. #1 SHIFT then #4 φ
40. #1 SHIFT then #5 GCD
41. #1 SHIFT then #7 MOD
42. #1 SHIFT then #8 INV m
43. #1 SHIFT then #9 b^cmod
44. #1 SHIFT then #10 →Q
45. #1 SHIFT then #11 NUM
46. #1 SHIFT then #12 EGYPT
47. #1 SHIFT then #13 FIB
48. #1 SHIFT then #14 DSUM
49. #1 SHIFT then #15 PERF?
50. #1 SHIFT then #16 π(n)
51. #1 SHIFT then #17 IP
52. #1 SHIFT then #18 FP
53. #1 SHIFT then #20 F↔D
54. #1 SHIFT then #21 x^2
55. #1 SHIFT then #22 yx
56. #1 SHIFT then #23 nCr
57. #1 SHIFT then #26 R↓
58. #1 SHIFT then #27 CLx
59. #1 SHIFT then #32 ⌫

Appendix B — Function finder (A–Z)

Function	Label	Keys
Addition +	+	#46
Backspace — delete the last digit typed	⌫	#32
Best fraction approximation	→Q	#10
ceil	CEIL	#1 → #17
Change sign (+/−) of the number being typed or the current result	CHS	#45
Chinese remainder (T mod Z, Y mod X)	CRT	#1 → #8
Clear stack	CLST	#1 → #27
CLx — clear X	CLx	#27
Collatz steps	COLTZ	#1 → #15
Combinations nCr	nCr	#23
Continued fraction	CF	#1 → #10
Count primes ≤ n	π(n)	#16
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	.	#44
Denominator	DEN	#1 → #11
Digit 0	0	#43
Digit 1	1	#38
Digit 2	2	#39
Digit 3	3	#40
Digit 4	4	#33
Digit 5	5	#34
Digit 6	6	#35

Digit 7	**7**	**#28**
Digit 8	**8**	**#29**
Digit 9	**9**	**#30**
Digit sum	**DSUM**	**#14**
Division ÷	**÷**	**#31**
Divisor count	**d(n)**	**#1 → #9**
Divisor sum	**σ**	**#1 → #4**
EEX — enter a power-of-ten exponent for the number in X	**EEX**	**#47**
Egyptian fraction	**EGYPT**	**#12**
ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value	**ENTER**	**#24**
Euler totient	**φ**	**#4**
Fibonacci number	**FIB**	**#13**
floor	**FLOOR**	**#1 → #16**
fp	**FP**	**#18**
Fraction / mixed / decimal display	**F ↔ D**	**#20**
Fraction bar during entry	**a/b**	**#19**
GCD of Y and X	**GCD**	**#5**
Integer quotient	**DIV**	**#1 → #7**
ip	**IP**	**#17**
LASTx — recall the X value used by the last operation	**LASTx**	**#1 → #26**
LCM of Y and X	**LCM**	**#1 → #5**
Mediant	**MEDNT**	**#1 → #12**
Multiplication ×	**×**	**#36**
Next prime	**NXTP**	**#3**
Numerator	**NUM**	**#11**
ON — power on	**ON**	**#6**

Perfect / abundant / deficient	PERF?	#15
Permutations nPr	nPr	#1 → #23
Previous prime	PRVP	#1 → #3
Primality test of X	PRIME?	#2
Prime factorization	FACTOR	#1 → #2
R↓ — roll the stack down	R↓	#26
RCL — recall a memory / variable	RCL	#42
Reciprocal x^{-1} (1/x)	1/x	#1 → #22
Reverse digits	DREV	#1 → #14
Round to the displayed precision	RND	#1 → #18
SETUP — angle unit, number format, display options	MODE	#1 → #20
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#1
Square root √	√x	#1 → #21
Square x^2	x^2	#21
STO — store the value into a memory / variable (press the target next)	STO	#37
Subtraction –	–	#41
Triangular number	TRI	#1 → #13
undo	UNDO	#1 → #32
x≷y — exchange X and Y	x ↔ y	#25
Y mod X	MOD	#7
Y^{-1} mod X	INV m	#8
y^x — raise Y to the power X (RPN: base first, then exponent)	yx	#22
Z^Y mod X	b^emod	#9

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Rational & number-theory RPN: the four-level stack holds exact fractions (math.js Fraction objects). Number-theory routines are original U-45 code; modular power uses BigInt.

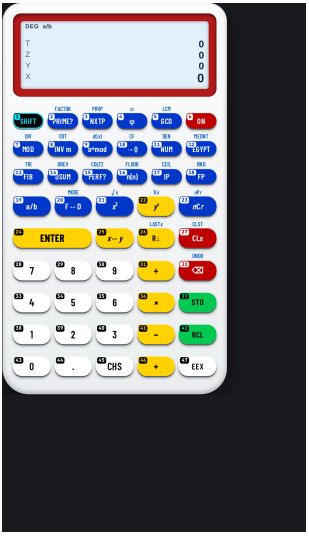
Results that cannot be exact (e.g. $\sqrt{2}$) are shown as decimals.

Planned (not yet working)

- Stack history view
- Undo
- Equation library
- Speech read-out
- Colour-blind key theme

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 SHIFT	**#2** PRIME? ⇧ FACTOR	**#3** NXTP ⇧ PRVP
#4 φ ⇧ σ	**#5** GCD ⇧ LCM	**#6** ON
#7 MOD ⇧ DIV	**#8** INV m ⇧ CRT	**#9** b^emod ⇧ d(n)
#10 → Q ⇧ CF	**#11** NUM ⇧ DEN	**#12** EGYPT ⇧ MEDNT
#13 FIB ⇧ TRI	**#14** DSUM ⇧ DREV	**#15** PERF? ⇧ COLTZ
#16 π(n) ⇧ FLOOR	**#17** IP ⇧ CEIL	**#18** FP ⇧ RND
#19 a/b	**#20** F ↔ D ⇧ MODE	**#21** x^2 ⇧ √x
#22 yx ⇧ 1/x	**#23** nCr ⇧ nPr	**#24** ENTER
#25 x ↔ y	**#26** R↓ ⇧ LASTx	**#27** CLx ⇧ CLST
#28 7	**#29** 8	**#30** 9
#31 ÷	**#32** ⌫ ⇧ UNDO	**#33** 4
#34 5	**#35** 6	**#36** ×
#37 STO	**#38** 1	**#39** 2
#40 3	**#41** −	**#42** RCL
#43 0	**#44** .	**#45** CHS
#46 +	**#47** EEX	

U-45 CALCULATORS · Calculator #19 · cut out and keep